# 120/2025-M

ചോദ്യ പുസ്തക അക്ഷര കോഡ് **A**

ചോദ്യ പുസ്തക ക്രമ നമ്പർ

ചോദ്യങ്ങളുടെ ആകെ എണ്ണം : 100

സമയം : 1 മണിക്കൂർ 30 മിനിട്ട്

പരമാവധി മാർക്ക് : 100

**ഉദ്യോഗാർത്ഥികൾക്കുള്ള നിർദ്ദേശങ്ങൾ**

1. ചോദ്യ കടലാസ്സ് നൽകുന്നത് ചോദ്യ പുസ്തക രൂപത്തിലാണ്. **A, B, C, D** എന്നീ നാല് അക്ഷര കോഡുകളിലുള്ള ചോദ്യ പുസ്തകങ്ങളാണ് നൽകുന്നത്.
2. ചോദ്യ പുസ്തകത്തിന്റെ മുകളിൽ ഇടത് വശത്ത് പ്രത്യേക കോളത്തിൽ ചോദ്യ പുസ്തക അക്ഷര കോഡ് അച്ചടിച്ചിട്ടുണ്ട്.
3. ഓരോ ഉദ്യോഗാർത്ഥിക്കും നൽകിയിട്ടുള്ള ചോദ്യ പുസ്തക അക്ഷര കോഡ് അവരുടെ പരീക്ഷാ ഹാളിലെ ഇരിപ്പിടത്തിൽ രേഖപ്പെടുത്തിയിരിക്കും.
4. നിങ്ങൾക്കനുവദിച്ചിരിക്കുന്ന അക്ഷര കോഡിൽ നിന്നും വ്യത്യസ്തമായ കോഡിലുള്ള ചോദ്യ പുസ്തകമാണ് ലഭിക്കുന്നതെങ്കിൽ അത് ഉടനടി ഇൻവിജിലേറ്ററുടെ ശ്രദ്ധയിൽപ്പെടുത്തേണ്ടതാണ്.
5. ചോദ്യ പുസ്തക ക്രമ നമ്പർ പുറംചട്ടയുടെ വലതു വശത്ത് മുകളറ്റത്തായി ചേർത്തിട്ടുണ്ട്. ക്രമ നമ്പർ ഇല്ലാത്ത ചോദ്യ പുസ്തകമാണ് നിങ്ങൾക്ക് ലഭിക്കുന്നതെങ്കിൽ അതേ അക്ഷര കോഡിലുള്ള ചോദ്യ പുസ്തകം മാറ്റി വാങ്ങുക.
6. ചോദ്യ പുസ്തകം വലതു മാർജിനിൽ മദ്ധ്യ ഭാഗത്തായി സീൽ ചെയ്തിരിക്കും. ഉത്തരമെഴുതാനുള്ള അനുമതി ലഭിച്ചാലല്ലാതെ ചോദ്യ പുസ്തകം തുറക്കാൻ പാടില്ല.
7. പരീക്ഷ തുടങ്ങിയാലുടൻ ഉദ്യോഗാർത്ഥി തനിക്ക് ലഭിച്ചിരിക്കുന്ന ചോദ്യ പുസ്തകത്തിൽ 100 ചോദ്യങ്ങളും ക്രമമായി ഉണ്ടോ എന്ന് പരിശോധിക്കേണ്ടതാണ്. ചോദ്യ പുസ്തകത്തിൽ അച്ചടിക്കാത്തതോ, കീറിയതോ, വിട്ടുപോയതോ ആയ പേജുകൾ ഉണ്ടാകില്ല; അഥവാ ഉണ്ടെങ്കിൽ അക്കാര്യം ഇൻവിജിലേറ്ററുടെ ശ്രദ്ധയിൽ കൊണ്ടുവരികയും അതേ അക്ഷര കോഡിലുള്ള പൂർണ്ണമായ ചോദ്യ പുസ്തകം മാറ്റി വാങ്ങുകയും ചെയ്യേണ്ടതാണ്. ഇത് ഏറ്റവും പ്രധാനപ്പെട്ടതാണ്.
8. ചോദ്യ പുസ്തകത്തിന്റെ അവസാനം ഒന്നും എഴുതാത്ത ഒരു കടലാസ് ചേർത്തിട്ടുണ്ട്. ഇത് ഉത്തരം കണ്ടുപിടിക്കുന്നതിനുള്ള കുറിപ്പുകൾക്കായി ഉപയോഗിക്കാവുന്നതാണ്.
9. **ഉത്തരങ്ങൾ രേഖപ്പെടുത്താൻ ആരംഭിക്കുന്നതിന് മുമ്പ് ഉത്തരക്കടലാസിന്റെ മറുപുറത്തുള്ള നിർദ്ദേശങ്ങൾ ശ്രദ്ധാപൂർവ്വം വായിക്കുക.**
10. ശരിയുത്തരം ഉൾപ്പെടെ ഓരോ ചോദ്യത്തിനും **(A), (B), (C), (D)** എന്ന നാല് ഉത്തരങ്ങൾ തന്നിരിക്കും. ശരിയുത്തരം തെരഞ്ഞെടുത്ത് ഒ.എം.ആർ. ഉത്തരക്കടലാസിൽ ബന്ധപ്പെട്ട ചോദ്യ നമ്പറിന് നേരെയുള്ള ശരിയുത്തരം സൂചിപ്പിക്കുന്ന കുമിള (ബബിൾ) മാത്രം നീലയോ കറുപ്പോ ബാൾ പോയിന്റ് പേന ഉപയോഗിച്ച് കറുപ്പിക്കുക.
11. **ഓരോ ശരിയുത്തരത്തിനും ഒരു മാർക്ക് ലഭിക്കുകയും ഓരോ തെറ്റുത്തരത്തിനും 1/3 മാർക്ക് നഷ്ടമാവുകയും ചെയ്യും. ഉത്തരം രേഖപ്പെടുത്താത്ത ചോദ്യങ്ങൾക്ക് മാർക്ക് നഷ്ടമാവുകയില്ല.**
12. പരീക്ഷാസമയം കഴിയുന്നതിന് മുമ്പോ, ഉത്തരക്കടലാസ് ഇൻവിജിലേറ്ററെ ഏൽപ്പിക്കാതെയോ ഒരു ഉദ്യോഗാർത്ഥിയും പരീക്ഷാഹാൾ വിട്ട് പുറത്തുപോകാൻ പാടില്ല.
13. നിർദ്ദേശങ്ങൾ കർശനമായി പാലിക്കേണ്ടതാണ്. പരീക്ഷയിൽ ക്രമക്കേട് നടത്തുകയോ അതിനുള്ള ശ്രമം നടത്തുകയോ ചെയ്യുന്ന ഉദ്യോഗാർത്ഥികളെ അയോഗ്യരായി പ്രഖ്യാപിക്കുന്നതാണ്.

VOCABSPEDIA

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Maximum : 100 marks

Time : 1 hour and 30 minutes

**1.** Which of the following statement/s about 'demonetisation' is/are true?

(i) Demonetisation is the process by which a nation's economic unit of exchange loses its legally enforceable validity.

(ii) Raghuram Rajan was the Governor of Reserve Bank at the time of demonetization of 2016.

(iii) On November 8, 2016, the central government has notified that the specified bank notes of denominations of the existing series of the value, of Rs. 500 and Rs. 1,000 shall cease to be legal tender with effect from November 9, 2016.

(iv) Demonetisation of 2016 was presented as an initiative of the central government targeted to address disparate evils plaguing the nation's economy including practices of hoarding black money and money laundering.

(A) (i), (ii), (iii) and (iv) (B) (i), (iii) and (iv)

(C) (i) and (iv) (D) (i), (ii) and (iv)

**2.** Which of the following statement/s about Han Kang who won the Nobel Prize for Literature in 2024 is/are incorrect?

(i) Han Kang began her career in 1993 as a poet, but has since written mainly novels and short stories.

(ii) Her most famous works are 'The Vegetarian', 'Human Acts' and 'We Do Not Part'.

(iii) 'The Vegetarian' was first published in 2017.

(iv) 'The Vegetarian' tells the story of a graphic designer who suddenly decides to stop eating meat.

(A) (i) and (iv) (B) (i) only

(C) (iii) only (D) (iv) only

**3.** Which of the following statement/s about Niti Ayog is/are incorrect?

(i) It started functioning since 01/04/2016.

(ii) Its full form is National Institution for Transforming India.

(iii) One of its declared objectives is to foster cooperative federalism.

(iv) Narendra Damodardas Modi was its first chairman.

(A) (i) only (B) (ii) and (iii)

(C) (iv) only (D) (i) and (iv)

**4.** Which of the following does not match?

(i) Lord Lytton — Vernacular Press Act

(ii) Lord Ripon — Local Self Government

(iii) Lord Linlithgow — Delhi Durbar

(iv) Lord Mayo — Census

(A) (i) and (iii) (B) (iii) only

(C) (i) and (iv) (D) (iv) only

**5.** To which of the following five year plans, the motto '*gharibi hatao*' (Removal of Poverty) is associated?

(A) Third Five Year Plan
(B) Fourth Five Year Plan
(C) Second Five Year Plan
(D) Fifth Five Year Plan

**6.** Arrange the following events occurred in 1919 in exact chronological sequence

(i) Formation of the 'Disorders Inquiry Committee'
(ii) Rowlatt Act
(iii) Jalianwala Bagh Massacre
(iv) Renunciation of Knighthood by Rabindranath Tagore

(A) (iii), (i), (ii), (iv)
(B) (ii), (iii), (iv), (i)
(C) (ii), (iii), (i), (iv)
(D) (iii), (i), (iv), (ii)

**7.** Which of the following does not match?

(i) Brahmananda Sivayogi — *Adwaitha chintha paddhathi*
(ii) Sri Narayana Guru — *Darsanamala*
(iii) Chattampi Swamikal — *Pracheena malayalam*
(iv) V.T. Bhattathirippad — *Markkudakkullile mahanarakam*

(A) (i) and (iv)
(B) (i) only
(C) (iv) only
(D) (iii) only

**8.** Which of the following statement/s about the 71st National Film Awards declared on 1st August 2025 is/are incorrect?

(i) '12th Fail', the Hindi language biographical film directed by Vidhu Vinod Chopra won the Best Feature Film Award.
(ii) Shah Rukh Khan and Vikrant Massey shared the Best Actor award for their performances in *Jawan* and *12th Fail* respectively.
(iii) *Ullozhukku* directed by Christo Tomy is adjudged the Best Malayalam Feature Film.
(iv) Vijayaraghavan has won the award for 'Best Actor in a Supporting Role' for his role in the Malayalam movie *Ullozhukku*.

(A) (iii) and (iv)
(B) (iii) only
(C) (iv) only
(D) (i) and (iii) only

**9.** What was the result of the cricket test match between India and England held at Edgbaston in July 2025?

(A) India beat England for 336 runs
(B) England won the match by 3 wickets
(C) India won the match by 3 wickets
(D) Ended in a draw

**10.** Which of the following statement/s about Marayoor is/are true?

(i) Marayoor is a small town situated in Idukki Taluk.
(ii) There exist some prehistoric rock paintings in Marayoor.
(iii) Marayoor was famous for its sandalwood forests.
(iv) A beautiful sight in Marayoor is the scenic Pambar river.

(A) All the above
(B) (i), (ii) and (iii)
(C) (ii), (iii) and (iv)
(D) (i), (iii) and (iv)

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**11.** Choose the appropriate question tag :
Please stop talking, _____?

(A) won't you? (B) will you?
(C) don't you? (D) do you?

**12.** Choose the word which is opposite in meaning to the given word :
Discrepancy

(A) consistency (B) respectful
(C) disrespectful (D) inconsistency

**13.** Pick out the right synonym for the word, awful from the options given :

(A) nice (B) fight
(C) friendly (D) terrible

**14.** Find the meaning of idiom, "beat around the bush".

(A) to agree completely (B) avoid saying something
(C) clear the forest (D) pamper one's baby

**15.** Complete the sentence with the correct preposition from the choices given :
We are very excited _____ our trip to Dubai next week.

(A) at (B) with
(C) over (D) about

**16.** Write the right form of the verb that agrees with the subject :
In his garden, the lights _____ on automatically.

(A) turns (B) turning
(C) is turning (D) turn

**17.** Write the comparative form of the adjective in the sentence given :
Michelle is _____ (tall) than her sister.

(A) tallest (B) tall
(C) taller (D) the tallest

**18.** Fill in the blank with suitable article :
This is _____ man I saw in Malaysia.

(A) an (B) a
(C) the (D) none

**19.** Choose the right one from the option to form the correct phrasal verb :
We couldn't take _____ our jackets in the whole trip. It was very cold everywhere.

(A) down (B) up
(C) off (D) in to

**20.** Select the word with correct spelling :
Mrs. Naveen introduced the new _____ to the PTA members.

(A) principle (B) prinsiple
(C) principal (D) prinsipal

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**21.** A wire of length 3 m and cross-sectional area 2 $mm^2$ is stretched by a force of 200 N. If Young's modulus of the material is $1.5 \times 10^{11}$ Pa, what is the extension produced :

(A) 0.2 mm (B) 2 mm

(C) 4 mm (D) 0.02 mm

**22.** Estimate the work done against the surface tension in blowing a soap bubble of diameter 2 cm. The surface tension of soap solution is $2.5 \times 10^{-2} \frac{N}{m}$ :

(A) $6.28 \times 10^{-5} J$ (B) $3.14 \times 10^{-5} J$

(C) $6.28 \times 10^{-3} J$ (D) $3.14 \times 10^{-3} J$

**23.** When two pipes having diameter (d), converge to form a pipe of diameter (D). What should be the relation between d and D, such that the velocity in the third pipe becomes one by fourth of that in each of the two pipes?

(A) $d = \frac{D}{2\sqrt{2}}$ (B) $d = \frac{D}{\sqrt{2}}$

(C) $d = \sqrt{2}\, D$ (D) $d = \frac{D}{2}$

**24.** Water from a tap emerges vertically downwards with an initial speed of $\sqrt{2.0}$ *m/s*. The cross-sectional area of the tap is $10^{-4} m^2$. Assume that the pressure is constant throughout the stream of water and that the flow is steady, the cross-sectional area of stream 0.10 *m* below the tap is (Given $g = 10\ m/s^2$) :

(A) $2 \times 10^{-4} m^2$ (B) $\sqrt{2} \times 10^{-4} m^2$

(C) $\frac{10^{-4}}{2} m^2$ (D) $\frac{10^{-4}}{\sqrt{2}} m^2$

**25.** A Carnot's engine whose temperature of the source is 500 K takes 2000 J of heat at this temperature and rejects 1500 J of heat to the sink. Calculate the efficiency of the engine :

(A) 20% (B) 25%

(C) 15% (D) 30%

**26.** 10 g at 0°C is mixed with 10 g of steam at 100°C. What is the final temperature of the mixture?

(A) 50°C (B) 25°C

(C) 100°C (D) 75°C

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**27.** An ice cube at 0°C is dropped on the ground melts to water at 0°C. If all the kinetic energy of the ice went into melting it, from what height did it fall. (Latent heat of fusion of ice = $3.35\times10^5 \frac{J}{Kg}$); $g$ = 10 m/s$^2$ ?

(A) $3.35 \times 10^5$ m (B) $3.35 \times 10^3$ m

(C) $3.35 \times 10^4$ m (D) $3.35 \times 10^6$ m

**28.** In Newton's rings experiment, diameter of the 5$^{th}$ and 25$^{th}$ rings were 0.562 cm and 0.762 cm. Find the radius of curvature of the plano convex lens if the wavelength of the light used was 662 nm. :

(A) 0.5 m (B) 0.05 m

(C) 0.75 m (D) 0.075 m

**29.** The core and cladding of a silica optical fiber have refractive index $n_1 = 1.55$ and $n_2 = 1.44$ . Calculate the acceptance angle of the optical fiber :

(A) 25° (B) 35°

(C) 45° (D) 50°

**30.** How many half period elements are there in a circular portion of $2\times10^{-2}\, m$ radius of a plane wavefront given that the wavelength is $500\times10^{-7}\, m$ and the distance of the point of observation of the wavefront is 2 meter?

(A) 8 (B) 2

(C) 4 (D) 6

**31.** Which of the following is not a possible value for the magnitude of the orbital angular momentum in hydrogen atom?

(A) $\sqrt{12\hbar}$ (B) $\sqrt{20\hbar}$

(C) $\sqrt{30\hbar}$ (D) $\sqrt{40\hbar}$

**32.** Obtain the Miller Indices of a plane parallel to $y$ and $z$ axis :

(A) (010) (B) (111)

(C) (001) (D) (100)

**33.** The photoelectric threshold wavelength of silver is 275 nm. Find the maximum kinetic energy of the ejected electron when the surface of the silver is illuminated with ultra violet light of wavelength 200 nm. (where $h = 6.63\times10^{-34} J/s$) :

(A) $1.98\times10^{-19}\, J$ (B) $1.98\times10^{-20}\, J$

(C) $1.98\times10^{-18}\, J$ (D) $1.98\times10^{-21}\, J$

**34.** Archeologists unearth charcoal from an ancient campfire and find its carbon-14 activity per unit mass to be 7.4% activity measured in living wood. Find the half-life period of the charcoal :

(A) 5370 $y$ (B) 5730 $y$

(C) 3750 $y$ (D) 3570 $y$

**35.** If the position of a particle is known with high accuracy, then according to Heisenberg's principle :

(A) Its momentum is exactly known

(B) Its momentum becomes highly uncertain

(C) Its momentum is zero

(D) Nothing can be predicted

**36.** Which of the following statements about para-hydrogen is correct?

(A) It is more stable at room temperature

(B) It has parallel nuclear spins

(C) It has antiparallel nuclear spins and total spin = 0

(D) It is unstable below 20 K

**37.** Which of the following is an example of intermolecular hydrogen bonding?

(A) Para-nitrophenol (B) Ortho-nitrophenol

(C) Both (A) and (B) (D) None

**38.** Peroxyacetyl nitrate (PAN), a constituent of photochemical smog, is formed from the reaction of nitrogen dioxide with :

(A) Sulfur dioxide (B) Ozone

(C) Volatile organic compounds (D) Carbon dioxide

**39.** Which of the following contains a coordinate covalent bond?

(A) $H_2$ molecule (B) $O_2$ molecule

(C) NaCl (D) $H_3O^+$ ion

**40.** Markovnikov's rule specifically applies to :

(A) Radical addition reactions

(B) Elimination reactions

(C) All nucleophilic substitutions

(D) Electrophilic additions where the electrophile is $H^+$

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41. The Lucas test is most useful for alcohols containing :

(A) More than six carbons

(B) Less than six carbons

(C) Aromatic rings only

(D) Polyhydroxy compounds only

42. The configuration of natural rubber is predominantly :

(A) cis-1, 4-polyisoprene

(B) trans- 1, 4-polyisoprene

(C) cis-1, 2-polyisoprene

(D) trans-1, 2-polyisoprene

43. Which of the following is currently used as an octane booster instead of TEL?

(A) Benzene and Toluene

(B) Cyclohexane and n-Heptane

(C) MTBE and ETBE

(D) Carbon tetrachloride and Chloroform

44. The expression $PV = (1/3)\ mNc^2$ represents :

(A) Van der Waals equation

(B) Kinetic gas equation

(C) Dalton's law of partial pressures

(D) Joule-Thomson effect

45. If the internal energy of a system increases by 40 J and at the same time the system does 10 J of work, the heat absorbed is :

(A) 30 J

(B) –50 J

(C) –30 J

(D) 50 J

46. Which of the following terms correctly describes the turnover frequency (TOF) of a catalyst?

(A) The number of substrate molecules adsorbed per unit surface area

(B) The number of catalytic cycles a site undergoes per unit time

(C) The reciprocal of the activation energy

(D) The rate of desorption of products per gram of catalyst

47. The Hittorf method for determining transport numbers is based on :

(A) The speed of ion migration

(B) The difference in ionic conductance

(C) The change in concentration of ions observed in the vicinity of the electrodes

(D) The emf of a concentration cell

48. In aliphatic ketones, the carbonyl stretching frequency is usually observed in the range :

(A) 1710 – 1720 $cm^{-1}$

(B) 1660 – 1700 $cm^{-1}$

(C) 1600 – 1650 $cm^{-1}$

(D) 1750 – 1800 $cm^{-1}$

OCABSPEDIA

**49.** Find out the algae which shows multiaxial thallus organization :

(A) *Batrachospermum*

(B) *Polysiphonia*

(C) *Oedogonium*

(D) *Volvox*

**50.** Which among the following is an adaptation of hydrophytes?

(A) Velamen tissue

(B) Sunken stomata

(C) Heterophily

(D) Salt glands

**51.** Which among the following is correct regarding potato tuber?

(A) Stem tuber

(B) Root tuber

(C) Bulbil

(D) Corm

**52.** The fruit type in arecanut is :

(A) Achene

(B) Capsule

(C) Drupe

(D) Follicle

**53.** Which among the following is not correct regarding secondary growth in dicot root?

(A) Cambial ring is completely secondary in origin

(B) The cambial ring is wavy initially

(C) Lenticels are not very prominent

(D) Periderm originates from cortical cells

**54.** Which among the following hormone can destroy weeds?

(A) Ethylene

(B) 2,4-D

(C) Abscisic acid

(D) Gibberellins

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**55.** Which is the final electron donor in non-cyclic photophosphorylation?

(A) Water
(B) ATP
(C) Cytochrome
(D) Ferredoxin

**56.** Find out the major drawback of transpiration pull theory from the following :

(A) It fails to explain the cohesion and adhesion of water molecules
(B) It fails to explain the role of stomata
(C) It could not explain how plants handle cavitation
(D) Mineral absorption mechanism is not explained

**57.** Edible palm oil is obtained from _____ part of the oil palm fruit.

(A) Endocarp
(B) Epicarp
(C) Endosperm
(D) Mesocarp

**58.** Find out the wrong statement regarding hemp :

(A) Hemp is obtained from bast fibers
(B) Sun hemp is *Crotalaria juncea*
(C) Hemp is obtained from the seed of *Corchorus capsularis*
(D) The botanical name of hemp is *Cannabis sativa*

**59.** A rain is considered as acid rain when the pH is less than :

(A) 5.6
(B) 6.2
(C) 7.0
(D) 6.0

**60.** Which among the following is/are correct?

(i) Olive ridley sea turtles are conserved in Gahirmatha Marine Wildlife Sanctuary
(ii) First marine national park in India is located at Gulf of Kutch, Gujarat
(iii) Jim Corbett national park is the first national park in India
(iv) African cheetahs were recently introduced in Kaziranga national park

(A) All of the above (i), (ii), (iii), (iv)
(B) Only (i) and (ii)
(C) Only (iii)
(D) (i), (ii) and (iii)

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**61.** Choose the incorrect statement :

(A) The concept of biodiversity hotspots was developed by Norman Myers

(B) The Great Banyan is a banyan tree (*Ficus benghalensis*) is conserved in Acharya Jagadish Chandra Bose Indian Botanic Garden, Kolkata

(C) The botanical garden in Kerala is famous for aquatic plant conservation is Malabar Botanical Garden and Institute for Plant Sciences

(D) Nilgiri Tahr is conserved in Aaralam wildlife sanctuary

**62.** Choose the correct statement from the following :

(i) Noise above 65 decibels (dB) is considered as noise pollution

(ii) Bromine can cause depletion of ozone layer

(iii) CFCs can cause ozone layer depletion

(iv) Deepwater Horizon oil spill, largest marine oil spill occurred in the Gulf of Mexico

(A) Only (iii) and (iv) correct

(B) All are correct (i), (ii), (iii) and (iv)

(C) Only (i), (iii) and (iv) are correct

(D) Only (iii) is correct

**63.** Angina pectoris means :

(A) Interference with the blood supply to the brain

(B) The volume of water retained in the vascular system regulated by hormones

(C) Chest pain occurs in the heart and often also in the left arm and shoulder

(D) The volume of blood pumped by each ventricle per minute

**64.** Which of the following clotting factor known as factor IV?

(A) Prothrombin

(B) Hageman factor

(C) Calcium

(D) Fletcher factor

**65.** A shift of oxygen hemoglobin dissociation curve to the right in response to increased blood carbondioxide and hydrogen ions are known as :

(A) Bohr effect

(B) Buffer effect

(C) Phosphate effect

(D) Shift effect

**66.** The portion of myofibril that lies between two successive '*Z*' disc is called :

(A) Sarcolemma (B) Sarcomere

(C) H-zone (D) Actomere

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**67.** Relaxin is secreted by :

(A) Pituitary gland

(B) Pineal gland

(C) Thyroid gland

(D) None of these

**68.** Most abundant immunoglobulin in human plasma is :

(A) IgG (B) IgM

(C) IgE (D) IgA

**69.** Which of the following statements are correct?

(i) The '*T*' wave on the ECG corresponds to the repolarisation of the ventricles

(ii) The AV node provides the only pathway for conduction of the depolarisation from atria to the ventricles

(iii) The second peak QRS in ECG, is produced by ventricular repolarisation

(iv) Baroreceptors located in the arch of aorta detects changes in arterial blood pressure

(A) All of the above

(B) Only (i), (ii) and (iv)

(C) Only (iii) is correct

(D) Only (ii) and (iv)

**70.** The domains of chromatin that are not expressed are called :

(A) Euchromatin

(B) Nucleosome

(C) Heterochromatin

(D) Demochromatin

**71.** Read the following statements :

(i) Bacterial cell wall consist of peptidoglycan

(ii) Down syndrome is a developmental defect produced by trisomy 21

(iii) Each strand of DNA is made up of repeating sugar and phosphate units joined by hydrogen bonds

(iv) Satellite DNAs are unique to an individual

(A) All the statements are correct

(B) All the statements are wrong

(C) Only (i) is correct

(D) Only (iii) is incorrect

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**72.** Identify Klinefelter's syndrome :

(A) 44A + XXY
(B) 44A + XO
(C) 45A + XX
(D) 45A + XXY

**73.** DNA duplex is coiled around a core of eight histone proteins forms :

(A) Scaffolden
(B) Solenoid
(C) Nucleosome
(D) Rosette

**74.** Cell mediated immunity is :

(A) One type of innate immunity
(B) Develops by circulating antibodies
(C) Activated through activated B-lymphocytes
(D) Acquired immunity activated by T-lymphocytes

**75.** Which of the following statement is incorrect?

(A) The resting membrane potential of large nerve fibers when not transmitting nerve signal is about –90 millivolts
(B) GABA is an example of neuro transmitter
(C) Broca's area in brain is associated with vision
(D) Dyslexia is the condition in which person may able to see words but not be able to interpret their meanings

**76.** Which among the following is not related to allergy?

(A) Histamine
(B) Mast cells
(C) Basophils
(D) Autograft

**77.** If $A$ and $B$ are two square matrices of same order, then $AB = I$ implies :

(A) Only $A$ is invertible
(B) Only $B$ is invertible
(C) Both $A$ and $B$ are invertible
(D) None of the above

**78.** Characteristic roots of a Hermitian matrix are :

(A) All equal
(B) All real numbers
(C) Conjugate pairs
(D) All complex numbers

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**79.** If $a,b,c$ are the distinct cube roots of unity (1), then which of the following is true?

(A) $a+b^2+c^3=0$

(B) $a+b+c=1$

(C) $abc=1$

(D) $abc=0$

**80.** Which of the following is the polar form of the complex number $1+i$ ?

(A) $2e^{\frac{\pi}{3}i}$

(B) $2e^{\frac{\pi}{4}i}$

(C) $\sqrt{2}e^{\frac{\pi}{3}i}$

(D) $\sqrt{2}e^{\frac{\pi}{4}i}$

**81.** Polar equation of the line bisecting the 1st and 3rd quadrants of the Cartesian plane is :

(A) $r\sin\theta=1$

(B) $r=\theta$

(C) $\theta=\dfrac{5\pi}{4}$

(D) $r=\dfrac{\pi}{4}$

**82.** The acute angle between the lines $x-2y-2=0$ and $3x+2y-6=0$ is :

(A) $\tan^{-1}8$ (B) $\tan^{-1}6$

(C) $\tan^{-1}\dfrac{3}{2}$ (D) $\tan^{-1}\dfrac{2}{3}$

**83.** What is the value of $\lim_{x\to1}\dfrac{1-x}{\log x}$ ?

(A) –1

(B) 1

(C) 0

(D) Does not exist

**84.** Which of the following is a point of inflection of the curve $y=3x^4-4x^3+1$ ?

(A) $(0,1)$ (B) $(1,0)$

(C) $\left(\dfrac{2}{3},\dfrac{8}{3}\right)$ (D) $\left(\dfrac{8}{3},\dfrac{2}{3}\right)$

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**85.** Order and degree of the differential equation $\frac{d^2y}{dx^2}+\left(\frac{dy}{dx}\right)^3+2xy^2=0$ are respectively :

(A) 1, 2 (B) 2, 1

(C) 2, 3 (D) 3, 2

**86.** Which of the following is an exact differential equation?

(A) $\sec y\left(x\tan y\,dy+dx\right)=0$

(B) $\sin y\left(x\cos y\,dy+dx\right)=0$

(C) $xydx+x^2dy=0$

(D) $xydy+y^2dx=0$

**87.** Angle between the vectors $3\hat{i}-4\hat{k}$ and $-3\hat{j}+4\hat{k}$ is

(A) $\tan^{-1}\frac{-16}{25}$ (B) $\sec^{-1}\frac{-16}{25}$

(C) $\sin^{-1}\frac{-16}{25}$ (D) $\cos^{-1}\frac{-16}{25}$

**88.** Which of the following vector is a solenoidal vector?

(A) $3x\hat{i}+(2y+z)\hat{j}-4\hat{k}$

(B) $(x+3y)\hat{i}+(y-2z)\hat{j}+(x-2z)\hat{k}$

(C) $(-2x+z)\hat{i}+3y\hat{j}+2z\hat{k}$

(D) $(x-2z)\hat{i}-(x+3y)j+(x+2y)\hat{k}$

**89.** In the cyclic group $Z_{12}$ (under addition modulo 12), Which of the following is a generator?

(A) 2 (B) 3

(C) 5 (D) 8

**90.** Which of the following is a possible number of elements in a field?

(A) 1 (B) 6

(C) 15 (D) 16

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**91.** ശരിയായ പദങ്ങൾ മാത്രം കണ്ടെത്തുക :

(i) നിഘണ്ടു

(ii) ഗരുഡൻ

(iii) മഠയൻ

(iv) സാമ്രാട്ട്

(A) എല്ലാം ശരി

(B) (i) മാത്രം ശരി

(C) (i)-ഉം, (ii)-ഉം, (iv)-ഉം ശരി

(D) (i)-ഉം, (iv)-ഉം ശരി

**92.** ശരിയായ വാക്യം കണ്ടെത്തുക :

(A) ഒരേ ഒരു കുട്ടി മാത്രമേ ഇനി വരാനുള്ളൂ

(B) ഒരേ ഒരു കുട്ടിയേ ഇനി വരാനുള്ളൂ

(C) ഒരേ ഒരു കുട്ടി മാത്രമേ ഇനിയും വരാനുള്ളൂ

(D) ഒരു കുട്ടി മാത്രമേ ഇനിയും വരാനുള്ളൂ

**93.** പമ്പ കടക്കുക എന്ന ശൈലിയുടെ അർത്ഥമെന്ത്?

(A) വിഷമത്തിലാവുക

(B) പമ്പാനദി മുറിച്ച് കടക്കുക

(C) പ്രയാസപ്പെടുത്തുക

(D) തോറ്റോടുക

**94.** 'ധനം' എന്ന പദത്തിന്റെ വിപരീതപദം താഴെ തന്നിരിക്കുന്നവയിൽ ഏതാണ്?

(A) അധനം (B) ആധനം

(C) ഋണം (D) ഋതം

**95.** 'മാംസഭുക്ക്' പര്യായപദം താഴെ തന്നിരിക്കുന്നവയിൽ ഏതാണ്?

(A) ആമിഷാശി

(B) മാംസികൻ

(C) വൈതംസികൻ

(D) കൗടികൻ

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96. തിരു + പാദം ചേർത്തെഴുതുമ്പോൾ :

(A) തിരുപ്പാദം

(B) തിരുപാദം

(C) തൃപാദം

(D) തൃപ്പാദം

97. 'ആദ്യന്തം' ഘടകപദങ്ങളാക്കുക :

(A) ആദ്യം അന്തം

(B) ആദിയും അന്തവും

(C) ആദ്യവും അന്തവും

(D) ആദിയായുള്ളതും അന്തമായിട്ടുള്ളതും

98. താഴെ തന്നിരിക്കുന്നവയിൽ സലിംഗബഹുവചനത്തിന് ഉദാഹരണം ഏത്?

(A) പശുക്കൾ

(B) കുട്ടികൾ

(C) മിടുക്കർ

(D) പണിക്കാർ

99. താഴെ തന്നിരിക്കുന്നവയിൽ പുല്ലിംഗത്തിന് ഉദാഹരണം ഏത്?

(A) കണ്ടിതു

(B) കണ്ടാൻ

(C) വന്നിതു

(D) വന്നാൾ

100. 'കേട്ടേ തീരൂ എന്നുള്ളത്' എന്നതിന്റെ ഒറ്റപ്പദം ഏതാണ്?

(A) വാചാലൻ

(B) സർവംസഹ

(C) ശ്രവ്യം

(D) ശ്രാവ്യം

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**FINAL ANSWER KEY**
**Question Paper Code:120/2025/M**
**Question Paper Medium:Malayalam**
**Category Code:449/2024,450/2024,784/2024,785/2024,786/2024,787/2024**
**Name Of Post & Department:Field Officer 449/2024, 450/2024, 784/2024, 785/2024, 786/2024, 787/2024**
**Date Of Test:25-Sep-2025**

| | QUESTION BOOKLET ALPHA CODE | | | |
|---|---|---|---|---|
| Q No. | A | B | C | D |
| 1 | B | A | D | A |
| 2 | C | B | B | C |
| 3 | A | D | A | A |
| 4 | B | B | C | C |
| 5 | D | A | A | B |
| 6 | B | C | C | C |
| 7 | A | A | B | A |
| 8 | C | C | C | B |
| 9 | A | B | A | D |
| 10 | C | C | B | B |
| 11 | B | D | D | C |
| 12 | A | B | D | C |
| 13 | D | D | C | C |
| 14 | B | D | C | C |
| 15 | D | C | C | B |
| 16 | D | C | C | A |
| 17 | C | C | B | D |
| 18 | C | C | A | B |
| 19 | C | B | D | D |
| 20 | C | A | B | D |
| 21 | B | D | C | B |
| 22 | A | B | C | C |
| 23 | A | C | A | D |
| 24 | D | C | B | D |
| 25 | B | A | C | A |
| 26 | C | B | D | B |
| 27 | C | C | D | B |
| 28 | A | D | A | A |
| 29 | B | D | B | A |
| 30 | C | A | B | D |
| 31 | D | B | A | B |
| 32 | D | B | A | C |
| 33 | A | A | D | C |
| 34 | B | A | B | A |
| 35 | B | C | D | C |
| 36 | C | D | B | B |
| 37 | A | D | A | D |
| 38 | C | B | C | B |
| 39 | D | A | B | C |
| 40 | D | C | D | A |
| 41 | B | B | B | B |
| 42 | A | D | C | C |
| 43 | C | B | A | A |
| 44 | B | C | B | C |
| 45 | D | A | C | D |
| 46 | B | B | A | D |
| 47 | C | C | C | B |
| 48 | A | A | D | A |
| 49 | B | C | B | D |
| 50 | C | D | A | C |

| | QUESTION BOOKLET ALPHA CODE | | | |
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| 52 | C | A | D | D |
| 53 | D | C | C | D |
| 54 | B | D | A | B |
| 55 | A | C | D | B |
| 56 | C | A | D | C |
| 57 | D | D | B | A |
| 58 | C | D | B | C |
| 59 | A | B | C | D |
| 60 | D | B | A | B |
| 61 | D | C | C | A |
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| 64 | C | D | B | A |
| 65 | A | A | C | C |
| 66 | B | B | D | D |
| 67 | D | C | A | C |
| 68 | A | D | C | D |
| 69 | B | A | D | C |
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| 72 | A | C | C | B |
| 73 | C | D | C | D |
| 74 | D | C | A | A |
| 75 | C | C | B | B |
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| 78 | B | C | A | A |
| 79 | C | A | B | D |
| 80 | D | A | B | B |
| 81 | C | B | A | C |
| 82 | A | B | D | D |
| 83 | A | A | B | D |
| 84 | B | D | C | B |
| 85 | B | B | D | C |
| 86 | A | C | D | D |
| 87 | D | D | B | C |
| 88 | B | D | C | A |
| 89 | C | B | D | A |
| 90 | D | C | C | B |
| 91 | B | D | A | C |
| 92 | B | C | D | A |
| 93 | D | A | C | B |
| 94 | C | D | A | D |
| 95 | A | C | B | B |
| 96 | D | A | D | B |
| 97 | C | B | B | D |
| 98 | A | D | B | C |
| 99 | B | B | D | A |
| 100 | D | B | C | D |

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'X' denotes deletion

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